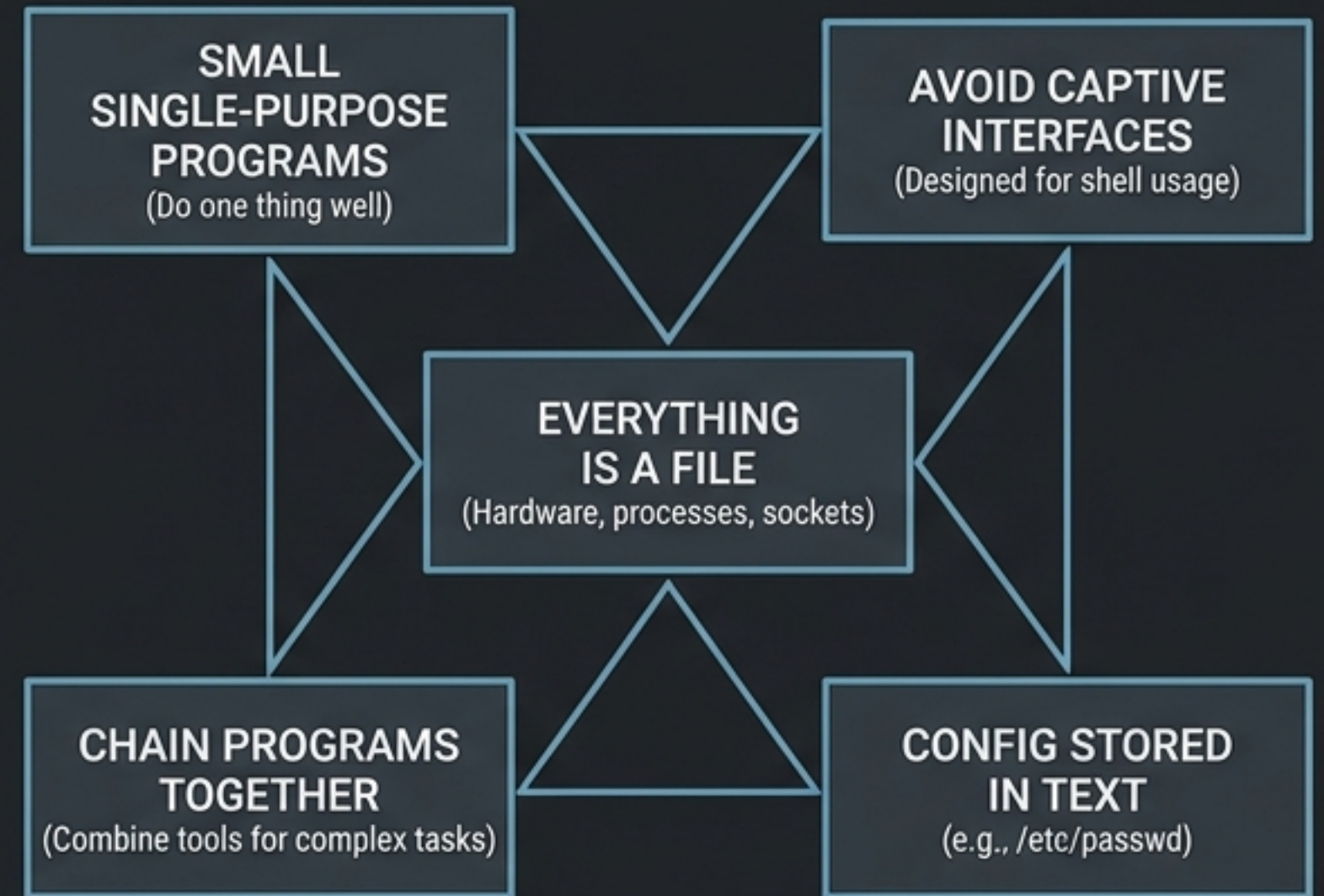
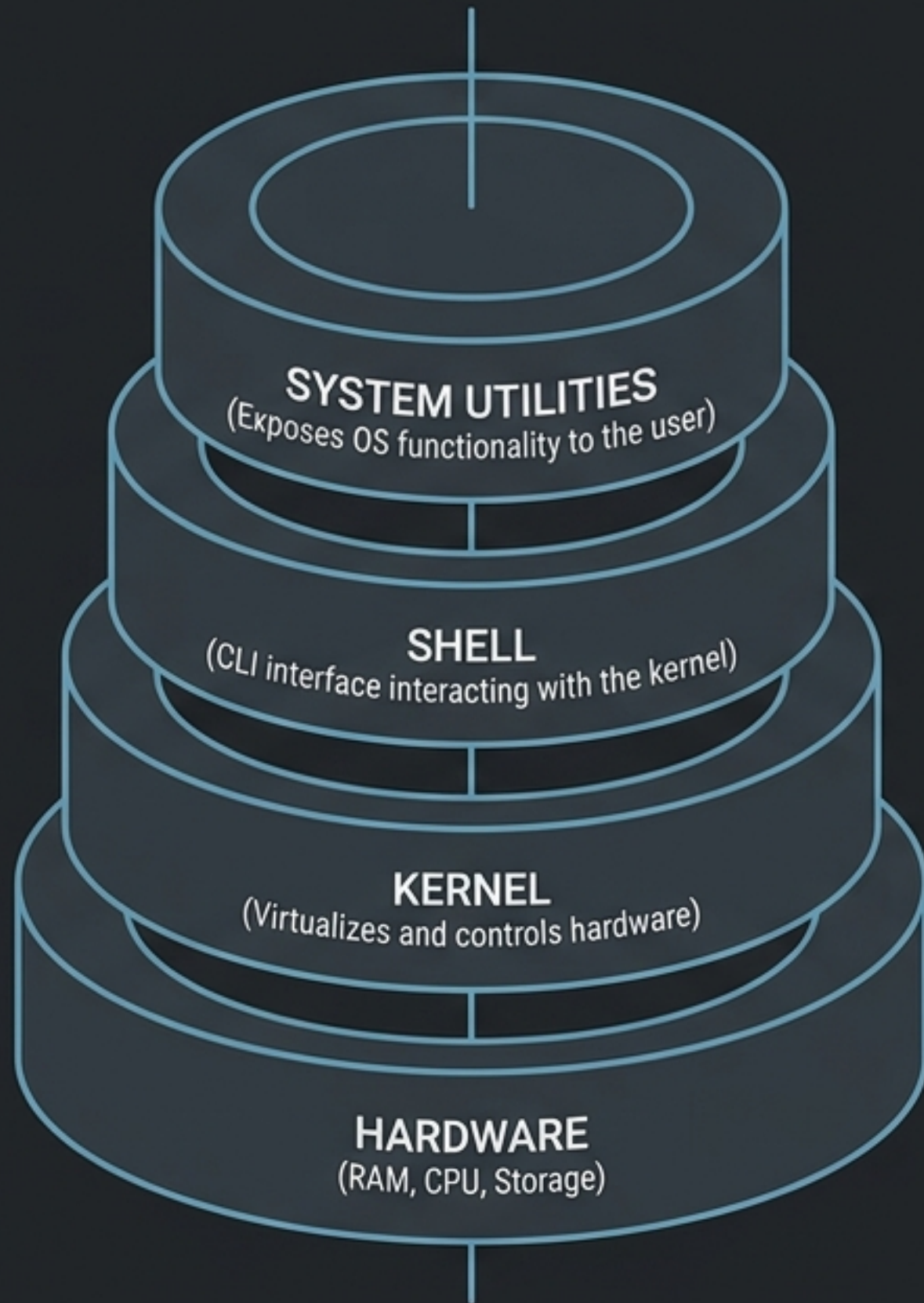


The Linux Operator's Blueprint

Fundamentals, Architecture, and Security Enumeration



Desktop/Server

Ubuntu
Fedora
CentOS
RHEL

Enterprise/Embedded

RHEL
Raspberry Pi OS

Debian is the foundation.
Known for stability and 5-year LTS.
Uses **Advanced Package Tool (apt)**.
Mastering Debian package management
is essential for operating Kali or Parrot
(the default OS for Hack The Box).

Solaris Contrast:

Linux uses **apt**,
ext4, and **sudo**.
Unix systems like
Solaris use IPS,
ZFS, and RBAC.

The Cybersecurity Arena

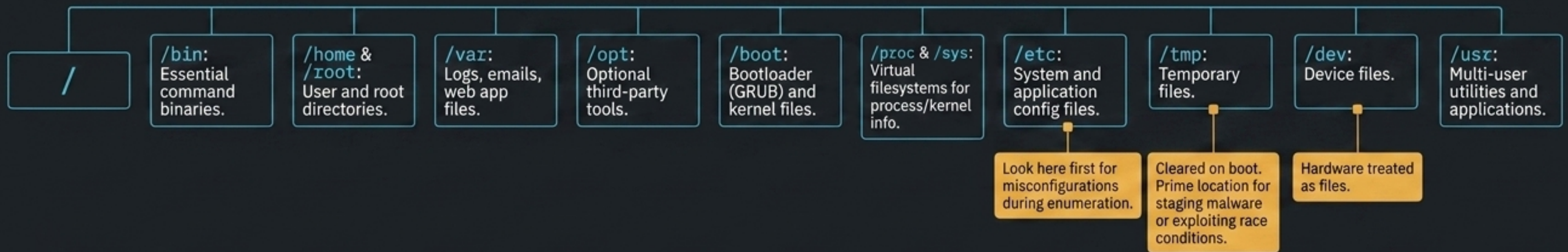
Kali Linux (Pen-testing standard)

BlackArch (Massive tool repo)

Parrot OS (Security/privacy focus)

Parrot OS

(Security/privacy focus)



Prompt Anatomy

`username@hostname:~$`

`\u` (username) `\h` (hostname) `\w` (working dir)

`$` (unprivileged user) vs `#` (privileged root)

A bare `$` or `#` means the `PS1` variable isn't set, common on compromised shells.

Situational Awareness: The Golden 5

`whoami + id`

Reveals current username, UID, GID, and critical group memberships (sudo, adm).

`uname -a / uname -r`

Prints kernel version (crucial for kernel exploit searching).

`hostname`

Current system name.

`ifconfig / ip`

Routing and interface addresses.

`netstat / ss`

Open sockets and connections.

Search Comparison Matrix

which : Finds executable paths. Best for checking if tools (python, nc) are installed.

locate : Fast database search (sudo updatedb). Broad, less granular.

find : Powerful, live filtered search.

```
find / -type f -name *.conf  
-exec ls -al {} \; 2>/dev/null
```

limits to files

runs commands
on results

The Help Workflow

Need deep
detail?

`man [tool]`

Need quick
flags?

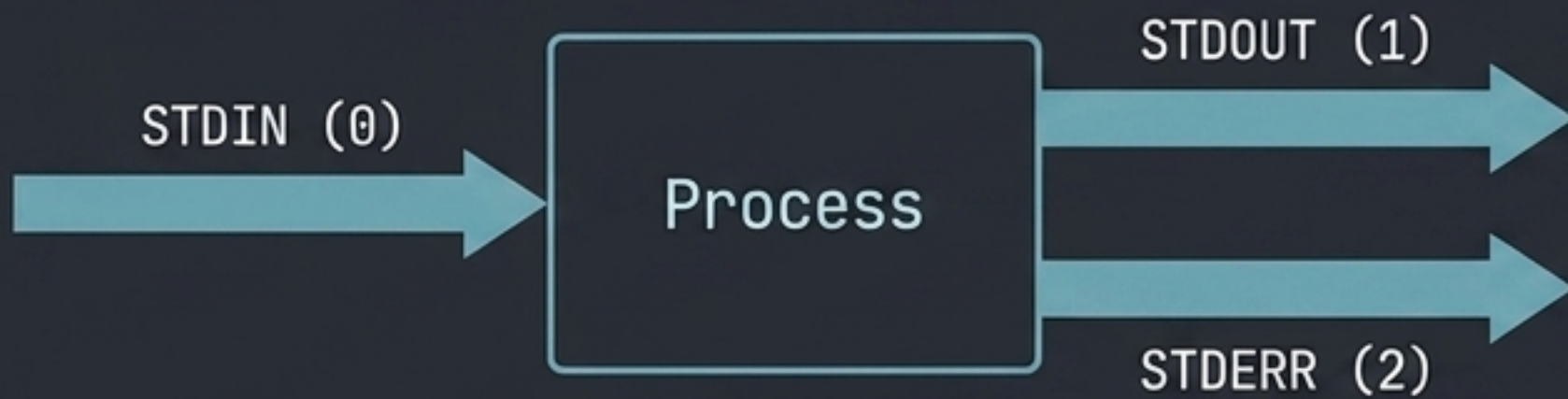
`[tool] --help or -h`

Know the goal
but not the tool?

`apropos [keyword]`

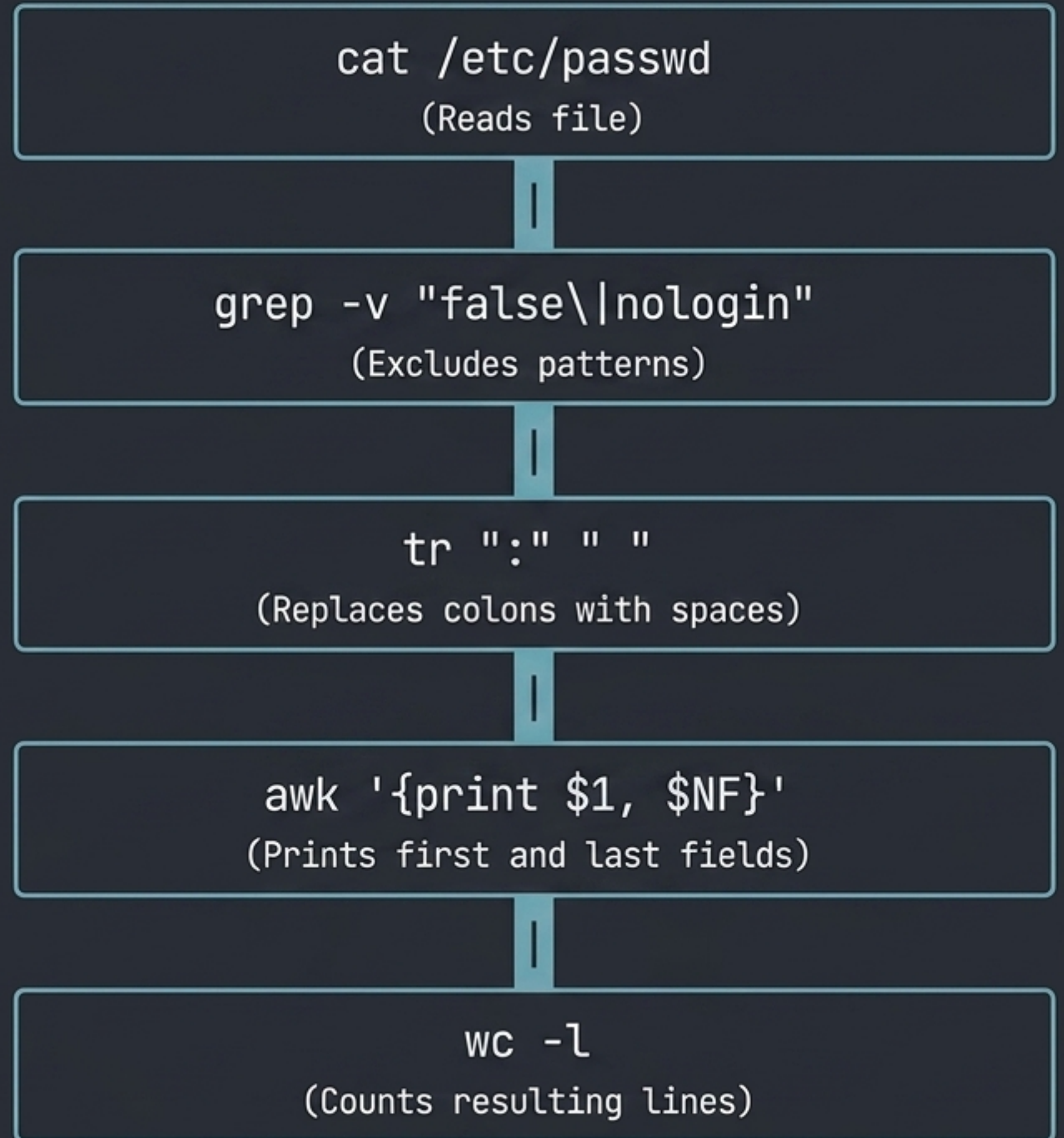
Emerald Tip:

Use explainshell.com for breaking down complex payloads found in the wild.



The Pen-Tester's Best Friend

`2>/dev/null` : Redirects STDERR (2) to the null device, suppressing permission denied errors during global searches.



The Anatomy of Permissions

d rwX r-X r-X

Directory Owner Group Others

Octal Notation: $r(4) + w(2) + x(1) = 7$ `chmod 754` = Owner `rwX`, Group `r-X`, Other `r--`

Privilege Escalation Vectors		
SUID (Set User ID)	SGID (Set Group ID)	Sticky Bit
Shown as 's' in owner execute. File runs with owner's privileges (often root). Check <code>GTF0bins!</code>	Shown as 's' in group execute. Runs with group privileges.	Shown as 't' on shared directories. Prevents users from deleting others' files.

Identity & Sudo

`/etc/passwd`

(Config: username, UID, GID, shell)



`/etc/shadow`

(Hashes, readable only by root)

Misconfigured `/etc/sudoers` is a primary privesc path.

Running '`sudo su`' or '`sudo cat /etc/shadow`' bypasses restrictions if poorly scoped.

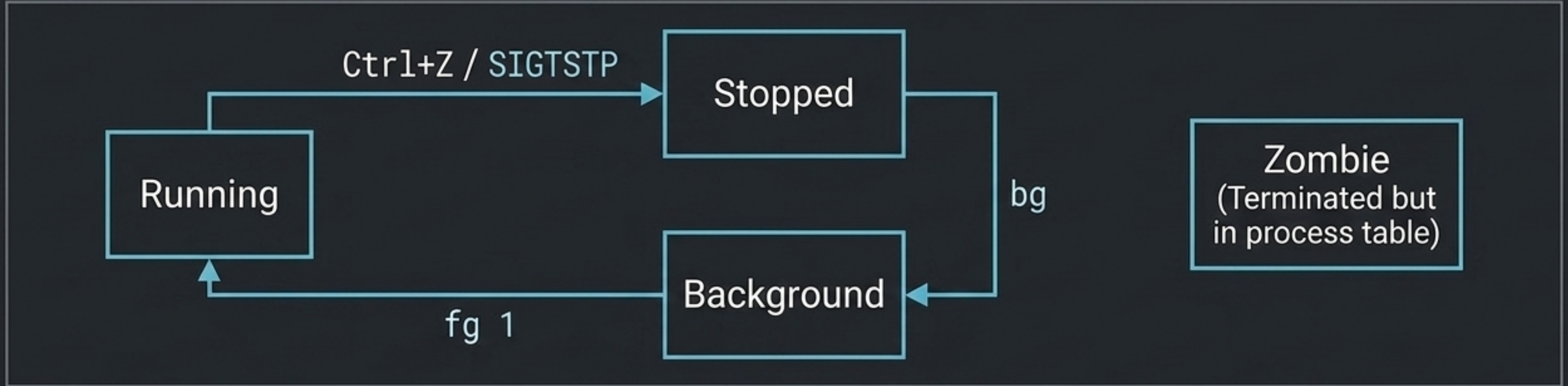
Package Management

`apt` : High-level Debian manager
(`apt update && apt install`).
Relies on `/etc/apt/sources.list`.

`dpkg` : Low-level tool.
Syntax: `dpkg -i package.deb`
(Used for manual, non-repo installs).

`git` : Essential for pulling external
pentesting tools (`git clone`).

Process Lifecycle



Critical Signals Matrix

1	HUP	(Terminal closed)
2	INT	(Ctrl+C interrupt)
15	TERM	(Graceful kill)
9	KILL	(Force kill, no cleanup)

Systemctl Core Commands

start	status
stop	enable
restart	disable

⚠ Note: 'enable' ensures persistence across reboots.

Task Scheduling & Persistence

Cron (The Legacy Standard)

```
* * * * * /path/to/script.sh
```

minute, hour, day-of-month, month, day-of-week

```
0 */6 * * * (Every 6 hours)
```

Amber Target Path Warning

/etc/crontab, /var/spool/cron/crontabs/,
/etc/cron.d/ (Highly abused for persistence)

Systemd Timers (The Modern Approach)

1. Timer File (OnBootSec=3min)
2. Service File
(ExecStart=/path/script.sh)
3. Activation (systemctl enable)

Synthesis

Cron is simpler and more widely abused;
Systemd offers robust logging via journalctl.

Common Network Services & Remote Access

Web & File Transfer

Fastest way to serve files:

```
python3 -m http.server 8000
```

wget

(Saves to disk)

curl

(Outputs to terminal)

SSH & VPN

Port 22. Standard for remote access:

```
ssh user@IP
```

Can encapsulate other traffic (X11 forwarding).

OpenVPN uses /etc/openvpn

NFS (Network File System)

Used for remote file sharing. Configured in /etc/exports

⚠️ `no_root_squash` allows a client root user to maintain root privileges on the remote share.

Remote Desktops

VNC: TCP 5900+, authenticated.

X11: TCP 6000–6009.

⚠️ Unencrypted by default, must be tunneled via SSH.

XDMCP: UDP 177, insecure.

The Container Landscape

Virtual Machines

Heavy, full isolated OS.

Docker

Application-focused, stateless, portable images.

LXC

System-focused, uses cgroups/namespaces.

⚠ Containers share the host kernel — they are not fully isolated. Misconfigurations lead to container escapes.

Filesystems & Inodes

Inode
(Metadata:
permissions,
timestamps)

Data Blocks
(File Data)

Inodes store metadata, not data or names (`ls -il`).

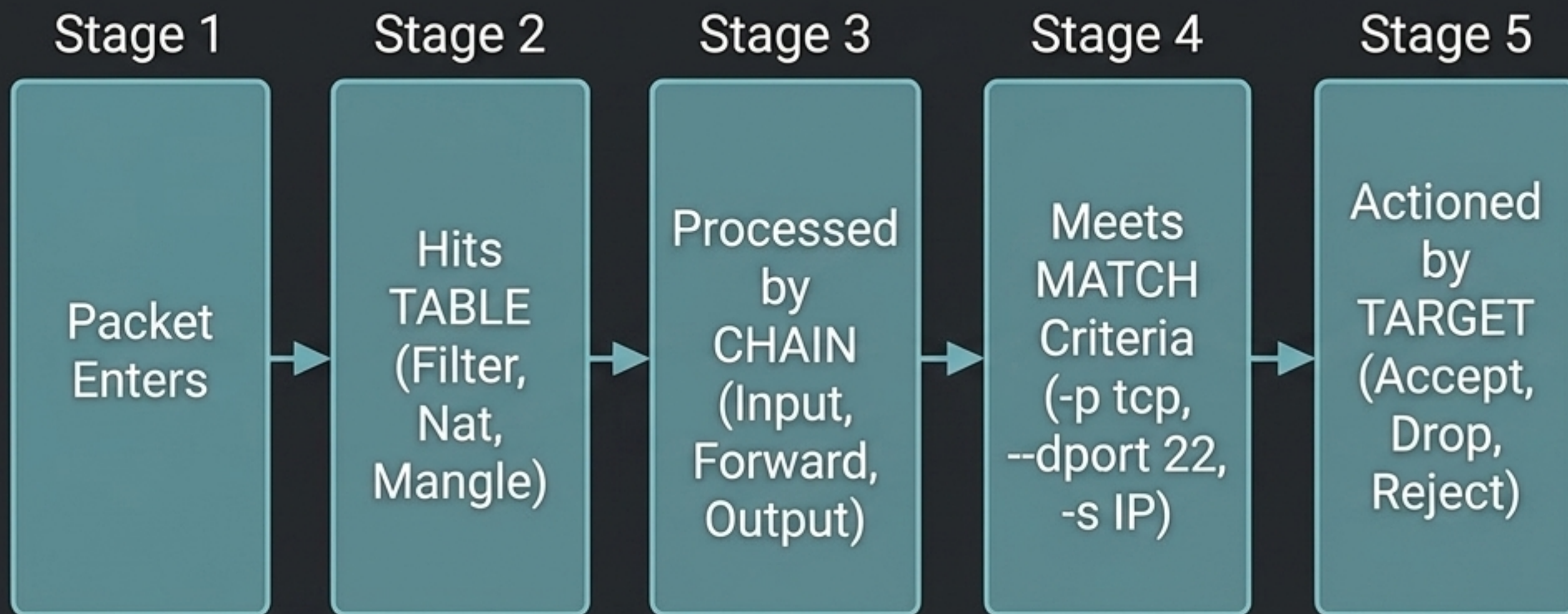
Quick Reference

Mounting

/etc/fstab controls boot mounts.

Always check `lsdf` before unmounting to prevent errors.

Hardening: Netfilter & Iptables



Cyan Rule Example

```
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

Hardening Alternatives

MAC Systems

SELinux
(kernel-level, labels every object)
vs AppArmor
(profile-based).

TCP Wrappers

/etc/hosts.allow
and /etc/hosts.deny.
Controls service access, not ports.
Not a firewall replacement.

Muscle Memory Shortcuts

<code>Ctrl+R</code>	Reverse history search
<code>Ctrl+C</code> / <code>Ctrl+Z</code>	Process control
<code>Ctrl+L</code>	Clear
<code>Ctrl+A</code> / <code>Ctrl+E</code>	Cursor jump to start/end
<code>Ctrl+U</code> / <code>Ctrl+K</code>	Erase to start/end

The Log Timeline

<code>/var/log/auth.log</code>	: Authentication, SSH, sudo usage (Primary security log).
<code>/var/log/syslog</code>	: General system events.
<code>/var/log/kern.log</code>	: Kernel and hardware crashes.
<code>/var/log/fail2ban.log</code>	: Brute force block events.

“In Linux, everything is a file. Master the text, master the system.”